

NETWORK COMMUNICATION PRESENTATION

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Task	Introduction + Application Layer
Communication Scenario	Opening a Secure Website
Main Responsibility	Explain how the communication starts at the Application layer

Introduction – What is TCP/IP?

TCP/IP (Transmission Control Protocol/Internet Protocol) is a set of communication protocols that allows devices to communicate and exchange data across networks, including the Internet. In simple terms, TCP/IP provides the rules that devices follow when data is prepared, addressed, transmitted, delivered, and received.

TCP and IP have different responsibilities:

- IP is responsible for logical addressing and routing. It uses IP addresses to identify the source and destination of packets and helps routers move packets toward the destination.
- TCP is a transport protocol that establishes a reliable connection between applications and helps ensure that data arrives correctly, in order, and without unnecessary duplication.
- Together, TCP/IP supports communication between applications running on different devices across interconnected networks.

What is the TCP/IP Model?

The TCP/IP model is a layered framework used to explain how data moves between devices over a network. Each layer has a specific responsibility and uses protocols that work with the other layers. For this presentation, the layers will be used to trace what happens when a user opens a secure website.

TCP/IP Layer	Main Responsibility	Examples for Secure Website
Application	Provides network services directly to the user/application.	DNS, HTTP/HTTPS, browser
Transport	Provides end-to-end communication between applications.	TCP, UDP; TCP port 443
Internet	Provides logical addressing and routing between networks.	IP, ICMP
Network Access	Moves frames across the local network and physical medium.	Ethernet/Wi-Fi, MAC addresses, switches

Selected Scenario – Opening a Secure Website

The selected scenario is a user opening a secure website, such as an HTTPS website, in a web browser. The communication begins when the user enters a website address, for example <https://icdfa.com>, and presses Enter.

The communication can be summarized as follows:

1. The web browser receives the website address from the user.

2. DNS is used to find the IP address associated with the website name.
3. The client prepares to communicate with the web server using HTTPS.
4. At the Transport layer, TCP establishes a connection using the three-way handshake.
5. The client uses an ephemeral source port while the web server normally listens for HTTPS traffic on destination port 443.
6. IP addresses identify the source and destination devices and allow routers to forward packets across networks.
7. Data is encapsulated as it moves down the TCP/IP layers before transmission.
8. At the destination, the data is decapsulated as it moves up the layers to the web server application.
9. The server sends responses back to the client, following the same layered process in reverse.

In other words,

1. User enters a secure website address in a browser.
2. The browser starts the communication process.
3. DNS helps translate the domain name into an IP address.
4. HTTP transfers web information between a browser and a web server.
5. HTTPS is HTTP communication protected using TLS.
6. HTTPS helps provide confidentiality, integrity and authentication.
7. DNS, HTTP and HTTPS are associated with the TCP/IP Application layer.
8. The TCP/IP Application layer maps mainly to the OSI Application, Presentation and Session layers.

Application Layer: How the Communication Starts

Role of the Web Browser

The web browser is the application through which the user requests the website. When the user enters a URL, the browser interprets the address and begins the process of locating and communicating with the web server.

DNS – Finding the Website's IP Address

The browser needs an IP address before it can communicate with the destination server. DNS (Domain Name System) translates a human-readable domain name, such as example.com, into an IP address. This allows the client to identify where the website can be reached.

- The user enters the website name.
- The client checks whether the DNS information is already available locally or in its cache.
- If necessary, a DNS query is sent to a DNS resolver.
- The resolver obtains the required IP address and returns it to the client.
- The browser can then use the destination IP address to begin the connection to the web server.

HTTPS – Secure Web Communication

HTTPS (HTTP Secure) is used when the browser communicates with a secure website. HTTPS protects web traffic by using TLS (Transport Layer Security) to provide encryption and help authenticate the server. The web application uses HTTP messages, while TLS protects the communication channel.

Communication starts at the Application layer when a user opens a secure website. This section covers the scenario, the role of the web browser, DNS, HTTP/HTTPS, the meaning and purpose of HTTPS, the TCP/IP Application layer, and its relationship to the OSI model.

Introduction to the Scenario

Opening a Secure Website

Scenario: A user wants to access a secure website using a web browser.

Example: <https://www.icdfa.edu.ng>

Basic communication journey:

User → Web Browser → Network → Web Server → Response

What happens?

- The user enters the website address.
- The browser starts the communication.
- DNS helps find the website's IP address.
- HTTPS is used to request the webpage securely.

In our scenario, a user wants to access a website. The user opens a web browser such as Google Chrome or Microsoft Edge and enters a website address, for example, <https://www.icdfa.edu.ng>.

Although this looks like a simple action, several things happen in the background before the webpage appears on the screen. The browser starts the communication, DNS helps find the server, and HTTPS is used for secure communication with that server.

In our presentation, we will follow this communication through the TCP/IP layers and explain the protocols involved.”

What Happens When the User Enters the Website Address?

From Website Name to Web Server

1. User enters URL: <https://www.icdfa.edu.ng>
2. Browser reads the URL.
3. DNS finds the IP address: www.icdfa.com → IP address.
4. Browser contacts the web server.
5. HTTPS request is sent.

6. Server sends a response.

Understanding DNS

DNS stands for Domain Name System. It translates a human-readable domain name, such as `www.example.com`, into an IP address that computers can use to locate the server.

A simple way to understand DNS is to compare it with the contacts on a phone. We save a person's name instead of memorising their phone number. When we select the name, the phone finds the number associated with it. DNS works in a similar way for website names and IP addresses.

TCP/IP Application Layer

The Application layer is the layer closest to the user. It provides network services used by applications such as web browsers.

Protocol	Meaning	Purpose
DNS	Domain Name System	Finds the IP address of a domain
HTTP	Hypertext Transfer Protocol	Transfers web content
HTTPS	Hypertext Transfer Protocol Secure	Protects web communication using TLS

HTTPS Security

HTTPS is HTTP communication protected using TLS. The "S" in HTTPS means "Secure."

Confidentiality: Helps prevent unauthorised people from simply reading the communication.

Integrity: Helps detect unauthorised changes to the data.

Authentication: Helps the browser verify the identity of the website/server.

HTTP vs HTTPS

HTTP is used to transfer web information between a browser and a server. HTTPS does the same basic job, but the communication is protected using TLS.

OSI Model Mapping

TCP/IP Model	OSI Model
Application	Application
Application	Presentation
Application	Session

The TCP/IP Application layer corresponds mainly to the OSI Application, Presentation and Session layers. The OSI Application layer provides network services directly to applications; the Presentation layer deals with data representation, encryption and compression; and the Session layer helps establish, manage and terminate communication sessions.

For HTTPS, TLS-related functions are often discussed in relation to the OSI Presentation/Session areas, although modern protocol stacks do not map perfectly one-to-one to the OSI model.